MCQ-10

Topic: Deep Learning – Neural Networks, Optimization, CNN & Object Detection

Student Name: _____ **Date:** _____ **Score:** _____

Instructions: Select the correct option by marking the checkbox ☐ next to it.

1. A single neuron has input $x = (1, 2)$, weights $w = (0.5, -1)$, and bias $b = 0.5$. It uses a ReLU activation. What is the output of the neuron in the forward pass?

   ☐ -1
   ☐ 0
   ☐ 1
   ☐ 0.5

2. Why is a non-linear activation function necessary in a deep neural network?

   ☐ It reduces the total number of parameters.
   ☐ It makes training faster on GPUs.
   ☐ Without it, stacked layers collapse into a single linear transformation.
   ☐ It normalizes the input features.

3. Which activation function has output range $(-1, 1)$ and is zero-centered?

   ☐ Sigmoid
   ☐ ReLU
   ☐ Softmax
   ☐ Tanh

4. The "dying ReLU" problem occurs because:

   ☐ ReLU outputs are bounded between 0 and 1.
   ☐ ReLU causes activations to explode in deep networks.
   ☐ Neurons with negative pre-activations output 0 and have zero gradient, so they may stop learning.
   ☐ ReLU is not differentiable at any point.

5. The vanishing gradient problem in deep networks is most strongly associated with which activation function, whose derivative has a maximum value of only 0.25?

   ☐ ReLU
   ☐ Sigmoid
   ☐ Leaky ReLU
   ☐ ELU

6. Backpropagation computes gradients of the loss with respect to every parameter primarily by applying which mathematical rule?

   - ☐ Bayes' theorem
   - ☐ Central limit theorem
   - ☐ Taylor series expansion only
   - ☐ Chain rule of calculus

7. A linear neuron computes $a = wx + b$ with $w = 1$, $b = 0$ and input $x = 2$. The target is $y = 5$ and the loss is $L = \frac{1}{2}(a - y)^2$. What is $\dfrac{\partial L}{\partial w}$?

   - ☐ 6
   - ☐ -6
   - ☐ 3
   - ☐ -3

8. What is the key difference between Stochastic Gradient Descent (SGD) and Batch Gradient Descent?

   - ☐ SGD estimates the gradient from a single sample (or a mini-batch) instead of the entire dataset, giving cheaper but noisier updates.
   - ☐ SGD does not use gradients to update weights.
   - ☐ SGD always converges in fewer epochs than Batch Gradient Descent.
   - ☐ SGD must use a fixed learning rate of 1.

9. In SGD with momentum, the update is $v \leftarrow \beta v + \nabla L, \quad w \leftarrow w - \eta v$. What is the main effect of the momentum term?

   - ☐ It assigns a separate learning rate to every parameter.
   - ☐ It applies weight decay to the parameters.
   - ☐ It accumulates an exponentially weighted history of gradients, damping oscillations and accelerating movement in consistent directions.
   - ☐ It normalizes the activations of each mini-batch.

10. The Adam optimizer combines which two ideas?

    - ☐ Momentum (first moment) and RMSProp-style adaptive scaling (second moment), with bias correction.
    - ☐ Dropout and batch normalization.
    - ☐ L_1 and L_2 regularization.
    - ☐ Newton's method and line search.

11. What is the main difference between Adam and AdamW?

    - ☐ AdamW removes the second moment estimate.
    - ☐ AdamW applies weight decay directly to the weights, decoupled from the adaptive gradient-based update.
    - ☐ AdamW uses a larger default value of β_1.

- ☐ AdamW can only be used with CNNs.

12. A convolutional layer has 32 filters of size 3×3 and is applied to an input with 3 channels. Each filter has one bias. How many trainable parameters does the layer have?

    - ☐ 288
    - ☐ 864
    - ☐ 896
    - ☐ 1024

13. An input of size 224×224 passes through a convolution with kernel 7×7, stride 2, and padding 3. what is the output spatial size?

    - ☐ 111×111
    - ☐ 112×112
    - ☐ 110×110
    - ☐ 113×113

14. What is the main purpose of a pooling layer in a CNN?

    - ☐ To increase the number of channels.
    - ☐ To add non-linearity through learnable weights.
    - ☐ To downsample feature maps, reducing computation and providing some translation invariance, without learnable parameters.
    - ☐ To compute the loss during training.

15. What distinguishes Fast R-CNN from the original R-CNN?

    - ☐ It runs the CNN separately on each of the $\sim$2000 region proposals.
    - ☐ It removes the need for bounding-box regression.
    - ☐ It predicts boxes on a grid in a single shot.
    - ☐ It runs the CNN once on the whole image and uses RoI pooling to extract features for each proposal.

16. What is the key innovation of Faster R-CNN?

    - ☐ A Region Proposal Network (RPN) that shares convolutional features and generates proposals using anchors.
    - ☐ Use of Selective Search to generate proposals.
    - ☐ Predicting boxes directly on an $S \times S$ grid.
    - ☐ Replacing NMS with soft voting.

17. Box $A = (0, 0, 4, 4)$ and Box $B = (2, 2, 6, 6)$ are given as (x_1, y_1, x_2, y_2). What is their IoU?

    - ☐ 0.25
    - ☐ 0.33
    - ☐ 0.50

- ☐ 0.143

18. In NMS with IoU threshold 0.5, the highest-scoring box B_1 is kept first. The IoU values of the other boxes with B_1 are $B_2 = 0.7$, $B_3 = 0.3$, $B_4 = 0.6$. Which boxes are suppressed by B_1?

    - ☐ B_2 and B_3
    - ☐ B_3 only
    - ☐ B_2 and B_4
    - ☐ B_2, B_3, and B_4

19. How does YOLO fundamentally differ from Faster R-CNN?

    - ☐ YOLO requires a separate region proposal stage.
    - ☐ YOLO is a single-stage detector that predicts bounding boxes and class probabilities in one forward pass.
    - ☐ YOLO does not use convolutional networks.
    - ☐ YOLO performs classification only, without localization.

20. In YOLOv1 with grid size $S = 7$, $B = 2$ boxes per cell, and $C = 20$ classes, what is the shape of the output tensor?

    - ☐ $7 \times 7 \times 25$
    - ☐ $7 \times 7 \times 50$
    - ☐ $7 \times 7 \times 30$
    - ☐ $14 \times 14 \times 30$